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3 July 1969
Superseding
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10 May 1966

MILITARY SPECIFICATION

BATTERY, SEA WATER

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers one type of primary sea water-activated battery, referenced herein as the "battery assembly."

2. APPLICABLE DOCUMENTS

2.1 Applicable documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

STANDARDS

FEDERAL

Fed. Test Method Standard No. 151	Metals; Test Methods

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MS 16625	Ring, Retaining
MS 29561	Packing, Preform

DRAWINGS

NAVAL ORDNANCE SYSTEMS COMMAND

NAVORD DL 1330510	Container, Shipping and Storage for Six Batteries, Sea Water 2526785

FSC 1355

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NAVORD Dwg 2526748	Afterbody
NAVORD DL 2526785	Sea Water Battery
NAVORD Dwg 2526883	Case, Battery
NAVORD Dwg 2571089	Battery Cell

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procuring functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Classification of requirements. The requirements for the battery assembly and cells are classified as follows:

REQUIREMENT	PARAGRAPH
General Requirements	3.2
Performance Requirements and Product Characteristics	3.3
Environmental Requirements	3.4
Accelerated Storage Life	3.5
Operational Service Life	3.6
Battery Cell Performance	3.7

3.2 General requirements

3.2.1 Data requirements. No data is required by this specification or approval documents referenced in Section 2 unless specified in the contract or purchase order. See 6.2.

3.2.2 First article sample. Unless otherwise specified in the contract or purchase order, a first article sample of the battery assembly shall be manufactured using the methods and procedures proposed for production. The sample will be tested as specified in Section 4 herein and is for the purpose of determining, prior to starting production, that the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

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3.2.3 Materials. Unless otherwise specified in the contract or purchase order, all materials and treatments of materials and parts used in the manufacture of the battery assembly shall be in accordance with the applicable documents listed on NAVORD DL 2526785.

3.2.4 Definitions. Definitions of terms appear in Section 6.

3.3 Performance requirements and product characteristics. Unless specified otherwise, the battery assembly shall satisfy the following performance requirements and product characteristics when tested as specified in Section 4.

3.3.1 Insulation and circuit resistance, before operation

3.3.1.1 Pin-to-pin. The insulation resistance between the pins of the electrical connector, with the arming wire installed, shall be greater than 20 megohms at a potential of 50 volts direct current.

3.3.1.2 Pin-to-case. The insulation resistance between the pins of the electrical connector and the battery case, with the arming wire removed, shall be greater than 20 megohms at a potential of 50 volts direct current.

3.3.1.3 Pin-to-pin. The circuit resistance between pins A and B of the electrical connector, with the arming wire removed, shall be 470 K ±94 K ohms.

3.3.2 Operating output

3.3.2.1 Low temperature, low salinity. The battery assembly, in the operating condition, shall be capable of producing a potential of 1.7 volts minimum across a 0.40 ohm resistive load within 1.0 second for a duration of 10 seconds minimum upon exposure to Type A electrolyte. The battery shall have been previously conditioned for 1 hour at 0° F. The electrolyte shall have been previously conditioned to a temperature of +32° Fahrenheit (F) (electrolyte type is defined in 6.3.9).

3.3.2.2 High temperature, high salinity. The battery assembly, in the operating condition, shall be capable of producing a potential of 2 volts minimum across a 0.40 ohm resistive load with 0.5 second for a duration of 10 seconds minimum upon exposure to Type B electrolyte. The battery shall have been previously conditioned for 1 hour at +140° F. The electrolyte shall have been previously conditioned to a temperature of +98° F (electrolyte is defined in 6.3.9).

3.3.2.3 Polarity. The polarity of the battery shall be as shown in Figure 1.

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3.3.3 Arming wire removal. There shall be no separation of the arming wire from the battery assembly when a 20 ±1 pound tension load is suddenly applied at an angle zero degrees with the top face of the battery assembly, as shown in Figure 2. The arming wire shall separate from the battery assembly when an 80 ±2 pound tension load is applied suddenly at an angle 60 degrees from the top face of the battery assembly, as shown in Figure 3. Removal shall not impair opening of the cap.

3.3.4 Pressure

3.3.4.1 High pressure, internal. The battery case and connector shall be capable of withstanding an internal pressure of 1250 pounds per square inch gage (psig) without leakage or permanent deformation.

3.3.4.2 High pressure, external. The battery assembly, with the arming wire in place, shall be capable of withstanding an external pressure of 1250 psig, applied to the top surface, without leakage or failure of the arming wire.

3.3.4.3 Internal leakage. The battery assembly shall exhibit a helium leak rate less than 1×10^{-5} atmospheric cubic centimeters per second (atm cc/sec) with an internal pressure of 15 ±1 psi above the testing environment.

3.3.5 Configuration

3.3.5.1 Dimensions. Dimensions of the battery assembly shall be in accordance with NAVORD DL 2526785.

3.4 Environmental requirements. Unless otherwise specified, the battery assembly shall be capable of conforming to the performance requirements and product characteristics of this specification after being exposed to the environmental conditions specified in this section. For definitions of ready, operating, and nonoperating see 6.3.1, 6.3.2, and 6.3.3.

3.4.1 Vibration, ready condition. The battery assembly, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the vibrations defined in Table I. The battery assembly shall be vibrated for 180 minutes each in its longitudinal axis and in two mutually perpendicular transverse axes at any temperature within the range of -45° F to +140° F. Vibration is defined in 6.3.4.

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TABLE I

VIBRATION SCHEDULE

Double Amplitude (in. ±10%) Minimum	Acceleration (g's ±10%) Minimum	Frequency (cps)
0.2	---	5 - 22
---	3.0	22 - 40
0.036	---	40 - 74
---	10	74 - 500

3.4.2 Acceleration, ready condition. The battery assembly, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the accelerations defined in Table II for a duration of 1 minute for each magnitude. Acceleration is defined in 6.3.5.

TABLE II

ACCELERATION SCHEDULE

Magnitude (g's ±10%)	Acceleration Direction (See Figure 1)
25	Longitudinal axis, Fwd
35	Longitudinal axis, Aft

3.4.3 Shock, ready condition. The battery assembly, mounted as shown in Figure 1, shall be capable of meeting the performance requirements after being subjected, in the ready condition, to the shock loads defined in Table III. Shock is defined in 6.3.6.

TABLE III

SHOCK SCHEDULE

Test Sequence	Peak Magnitude (g's ±10%)	Duration (ms)	Impact Axis	Direction of Deceleration Force	Number of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3

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TABLE III (Cont'd)

SHOCK SCHEDULE

Test Sequence	Peak Magnitude (g's ± 10%)	Duration (ms)	Impact Axis	Direction of Deceleration Force	Number of Impacts
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

3.4.4 Temperature and humidity

3.4.4.1 Nonoperating

(a) Temperature. The battery assembly shall be capable of meeting the performance requirements subsequent to withstanding, in the nonoperating condition, environmental temperatures between -45° F and +140° F. Temperature is defined in 6.3.7.

(b) Humidity. The battery assembly shall be capable of meeting the performance requirements subsequent to withstanding, in the nonoperating condition, an environment of at least 95 percent relative humidity at +110° F for a minimum period of 8 hours. Humidity is defined in 6.3.8.

3.4.4.2 Ready, operating. The battery assembly, immediately after being conditioned for 1 hour at any temperature between 0° F and +140° F, shall be capable of meeting the performance requirements at any environmental temperature between 0° F and +140° F.

3.4.4.3 Marine atmosphere. The battery assembly, installed in a fixture in the ready condition, Figure 1, shall be subjected to a simulated sea atmosphere in accordance with Federal Test Method Standard No. 151, Method 811.1, for 50 hours.

3.5 Accelerated storage life

3.5.1 Packaged and packed condition. The battery assembly, in the packaged and packed condition, shall be capable of withstanding an environmental temperature of +140° F for a period of 30 days. Thereafter, the battery assembly shall be capable of operating in accordance with 3.3.2.1 or 3.3.2.2. This requirement satisfies a five year storage life in the packaged and packed condition.

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3.5.2 Ready condition. The battery assembly, in the ready condition, shall be capable of withstanding an environmental temperature of +140° F for a period of 14 days. Thereafter the battery assembly shall be capable of operating in accordance with 3.3.2.1 or 3.3.2.2. This requirement satisfies a one year storage life in the ready condition.

3.6 Operational service life. The battery assembly shall have an operational service life of one duty cycle.

3.7 Battery cell performance. The battery cell, when tested separate from the battery assembly, shall meet the following requirements:

3.7.1 Insulation resistance. The insulation resistance of the battery cell, between the two leads, shall be greater than 20 megohms when measured at a potential of 50 volts direct current.

3.7.2 Operation. The battery cell, in the operating condition, shall be capable of producing a potential of 2.5 volts minimum across a 0.40 ohm resistive load within 1.0 second for a duration of 10 seconds minimum upon exposure to Type A electrolyte. The battery cell shall have been previously conditioned for 1 hour at 0° F. The electrolyte shall have been previously conditioned to a temperature of +33° F (electrolyte is defined in 6.3.9).

3.8 Workmanship. The workmanship and finish shall be of sufficiently high grade to insure satisfactory operation consistent with the requirements of this specification. The assembly shall be thoroughly clean and free of excess materials, chips, and loose or spattered foreign materials. The standards of workmanship exhibited in any approved first article sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection of the battery assembly and cell shall be classified as follows:

(a) First article inspection (Subsection 4.4)

(b) Quality conformance inspection (Subsection 4.5)

(c) Periodic inspection (Subsection 4.6)

4.3 Sampling

4.3.1 First article sample. A first article sample of 32 battery assemblies manufactured in accordance with 3.2 shall be subjected to first article tests listed in Table IV and detailed in 4.4 through 4.4.4.15. Testing shall be done at an activity designated by the procuring activity. Acceptance of the first article sample shall be based on no defects in the sample. See 4.4.6. Further production of the battery assemblies by the supplier, prior to approval of the first article sample, shall be at the supplier's risk.

4.3.2 Quality conformance inspection sample. The sampling plans and procedures set forth in MIL-STD-105 shall be used in the determination of acceptability of the battery assembly, inspected as specified in 4.5.7.6 through 4.5.7.8. In addition, a quantity from a production lot of battery cells submitted for quality conformance testing in accordance with the sample plans and procedures set forth in MIL-STD-105, shall be subjected to the tests detailed in 4.5.8.1 and 4.5.8.2. A production lot of battery cells shall consist of a homogeneous lot produced to the same design, from the same materials, and to the same manufacturing process, and shall yield sufficient battery cells to constitute a minimum of three periodic battery assembly lots. Each battery lot shall consist of cells from the same lot.

4.3.3 Periodic production sample. Periodic production samples of battery assemblies shall be selected from units that have satisfactorily passed the quality conformance inspections of 4.5. These samples shall be subjected to the tests of 4.6 at an activity designated by the procuring activity. The periodic lot shall consist of 150 units maximum from a single lot of accepted battery assemblies as defined in 4.3.2. The periodic sample shall consist of 3 units randomly selected from the periodic lot.

When a defective unit is found in any group, acceptance of all units on hand shall be withheld and the contractor shall:

(a) Determine the cause of the defect.

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(b) Take the necessary actions to correct or eliminate the defect from all units on hand and, if so directed by the procuring activity, from those units previously shipped.

(c) Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

(d) Submit to the procuring activity for approval the proposed re-test program to verify that the corrective action eliminated the defect. Return to Step 1 and proceed as indicated when inspection is resumed.

4.4 First article inspection

4.4.1 Tests and examinations. The first article sample of 32 battery assemblies shall be inspected in accordance with Table IV, with 4 battery assemblies being tested in accordance with each of the 8 test sequences, A through H.

TABLE IV

FIRST ARTICLE TEST SEQUENCE

Tests	Paragraph Reference	Test Sequence							
		A	B	C	D	E	F	G	H
Inspection	4.4.4.1	1	1	1	1	1	1	1	1
Vibration, Ready Condition	4.4.4.2	2							
Acceleration, Ready Condition	4.4.4.3		2						
Shock, Ready Condition	4.4.4.4		3						
Temperature and Humidity, Nonoperating Condition	4.4.4.5	3		2					
Marine Atmosphere, Ready Condition	4.4.4.6						2		
Vibration, Packaged and Packed	4.4.4.7				2				
Shock, Packaged and Packed	4.4.4.8				3				
High Pressure, External	4.4.4.9					2	3		
Internal Leakage	4.4.4.10	4	4	4		3	4		
Arming Wire Pull Test	4.4.4.11		5	5	4				
Accelerated Storage Life, Packaged and Packed Condition	4.4.4.12							2	
Accelerated Storage Life, Ready Condition	4.4.4.13								2
Low Temeprature, Low Salinity Operation	4.4.4.14	5	6		5			3	3
High Temperature, High Salinity Operation	4.4.4.15			6		4	5		

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4.4.2 Test conditions. Unless otherwise specified, the battery assemblies and cells shall be subjected to the tests in 4.4.1 under the following conditions:

(a) Temperature: Room ambient +32° F to +98° F

(b) Altitude: Normal ground atmospheric pressure

(c) Vibration: None

(d) Humidity: Room ambient

4.4.3 Test equipment and setup. In addition to the items listed in 4.5.6, test equipment is required which provides the following characteristics:

4.4.3.1 Vibration equipment with temperature chamber. The vibration equipment with temperature chamber shall have the following characteristics:

(a) Frequency range, logarithmic sweep: 5 to 500 cycles per second (cps)

(b) Test specimen: 10 pounds

(c) Acceleration and displacement (ready condition):

5 to 22 cps	0.2 inch
22 to 40 cps	3.0 gravitational units (g's)
40 to 74 cps	0.036 inch
74 to 500 cps	1g

(d) Acceleration (packed and packaged):

5 to 60 cps	1g

(e) Automatic cycling time range: 0 to 15 minutes (optional)

(f) Temperature chamber range: -45° F to +140° F accuracy to ±5° F

(g) Interior size: 8 inches x 8 inches x 46 inches minimum

4.4.3.2 Acceleration equipment. The acceleration equipment shall have the following characteristics: 35 g's with a 10 pound test specimen.

4.4.3.3 Shock equipment. The shock equipment shall be capable of accepting a 10 pound test specimen and shall provide ranges of:

(a) 300 g's for 0.5 millisecond (ms) to 1.5 ms

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(b) 60 g's for 8.0 ms to 10.0 ms

(c) 45 g's for 18.0 ms to 20.0 ms

4.4.3.4 Temperature and humidity conditioning equipment. One or more equipments may be utilized to provide the following characteristics:

(a) Temperature: -45° F to +140° F; with an accuracy to ±5° F

(b) Relative humidity: 95 percent minimum at +110° F; with an accuracy to ±5° F

(c) Interior size: 8 inches x 8 inches x 48 inches minimum

This equipment may be used for both nonoperating and ready/operating temperature (and humidity) tests.

4.4.4 Test procedures, battery assembly

4.4.4.1 Inspection. The battery assemblies shall be subjected to the inspections detailed in 4.5.7.1 through 4.5.7.6.

4.4.4.2 Vibration, ready condition. The battery assembly, in the ready condition, shall be mounted in a manner equivalent to that shown in Figure 1. The resonant modes in each of the 3 axes shall be determined by varying the frequency of applied vibration slowly from 5 cps to 500 cps. The battery assembly shall then be subjected to resonance and cycling vibration in each axis, in conformance with the times shown in Table V. If more than 4 resonances are encountered in any axis, the 4 most severe resonances shall be chosen for the test. The resonant and cycling period, for each axis, shall be divided into 3 equal parts, which are conducted at -45° F ±5° F, room temperature, and +140° F ±5° F respectively. For cycling, a logarithmic frequency sweep rate shall be used if available; otherwise, other cycling rates of frequency change may be used. The sweep rate shall be such that 15 minutes actual vibration time is required to proceed from the lower to higher and back to the lower extremity of frequency range in accordance with the accelerations or double amplitudes defined in Table VI.

TABLE V

VIBRATION TEST SCHEDULE

Number of Resonances	Total Vibration Time at Resonance *	Cycling Time
0	---	3 hrs
1	30 min	2½ hrs

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TABLE V (Cont'd)

VIBRATION TEST SCHEDULE

Number of Resonances	Total Vibration Time at Resonance *	Cycling Time
2	1 hr	2 hrs
3	1½ hrs	1½ hrs
4	2 hrs	1 hr

*30 Minutes Each Resonance

TABLE VI

VIBRATION TEST SCHEDULE

Frequency (cps)	Double Amplitude (in. ±10%) Minimum	Acceleration (g's ±10%) Minimum
5 - 22	0.2	---
22 - 40	---	3
40 - 74	0.036	---
74 - 500	---	10

4.4.4.3 Acceleration, ready condition. The battery assembly, in the ready condition, shall be subjected to the accelerations defined in Table VII after being secured to an acceleration equipment (see Figure 1). Acceleration shall be measured at the approximate geometric center of the battery assembly.

TABLE VII

ACCELERATION SCHEDULE

Magnitude (g's ±10%)	Acceleration Direction (See Figure 1)	Time
25	Longitudinal axis, Fwd	1 minute
35	Longitudinal axis, Aft	1 minute

4.4.4.4 Shock, ready condition. The battery assembly, in the ready condition, shall be attached to a shock test machine (see Figure 1). Shock magnitudes and durations shall be measured by an accelerometer

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mounted on the mounting fixture. Tests on the battery assembly shall be performed in accordance with Table VIII in the sequence shown.

TABLE VIII

SHOCK TEST SCHEDULE

Test Sequence	Peak Magnitude (g's ±10%)	Duration (ms)	Impact Axis	Direction of Deceleration Force	Number Of Impacts
1	300	0.5 to 1.5	Longitudinal	Aft	3
2	85	0.5 to 1.5	Transverse A	Optional	3
3	85	0.5 to 1.5	Transverse B	Optional	3
4	45	18.0 to 20.0	Longitudinal	Fwd	3
5	60	8.0 to 10.0	Longitudinal	Aft	3
6	60	8.0 to 10.0	Longitudinal	Fwd	3
7	60	8.0 to 10.0	Transverse A	Optional	3
8	60	8.0 to 10.0	Transverse B	Optional	3

4.4.4.5 Temperature and humidity, nonoperating condition. Eight first article battery assemblies shall be subjected, in the nonoperating condition, to temperature and humidity cycling in the following manner: The specimen shall be placed in a temperature chamber which has been stabilized at an environmental temperature of -45° F +0° -5° F. The specimen shall remain in the environment for 6 hours. (As an alternative procedure, the specimen may be placed in the chamber, with the chamber at room ambient temperature. Chamber temperature shall then be reduced to -45° F +0° F -5° F within 2 hours maximum. The specimen shall thereafter remain in the chamber for 4 hours.) The specimen shall then be subjected to an environmental temperature of +140° F +5° F -0° F. Maximum transition time from the low temperature to the high temperature shall be 2 hours. The specimen shall remain in the high temperature environment for 4 hours. This cycle shall be performed 3 times; maximum transition time shall be 2 hours. The specimen shall subsequently be subjected to an environment of +110° F +5° F -0° F at a minimum relative humidity of 95 percent for a minimum of 8 hours. Thereafter, 4 battery assemblies shall be operated in accordance with 4.4.4.14 and 4 battery assemblies shall be operated in accordance with 4.4.4.15.

4.4.4.6 Marine atmosphere. The battery assembly, installed in a fixture, Figure 1, shall be subjected to a simulated sea atmosphere in accordance with Federal Test Method Standard No. 151, Method 811.1. A 20 percent salt solution shall be used. A time duration of 50 hours is required.

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4.4.4.7 Vibration, packaged and packed. The battery assembly, packaged and packed in accordance with Section 5, shall be vibrated in such a manner as to determine the frequency producing maximum transmissibility within the frequency range of 5 to 60 cps. The container shall be fastened, in suitable fashion, to the table of a vibration testing device producing an essentially sine wave motion. The maximum permissible motion freedom shall be approximately 1/32 inch. The specimen shall then be vibrated for 1 hour at the frequency producing maximum transmissibility and at an input acceleration of 1 g. This test shall be applied to each of the three principal orthogonal axes.

4.4.4.8 Shock, packaged and packed. The battery assembly, packaged and packed in accordance with Section 5, shall be subjected to free fall drop tests. The battery assembly in its container shall be dropped from a height of 36 inches. A total of four edgewise drops shall be made. Two drops each shall be made on the bottom and top edges at points 180 degrees apart. In addition, flat drops shall be made, one each, on the bottom and on the side. The drops shall be made on an impact surface of essentially nonyielding material such as stone or concrete. The complete series of drops shall be performed under each of the following conditions: room temperature, within 10 minutes after the container has been stabilized at -45° F ±5° F, and within 10 minutes after the container has been stabilized at +140° F ±5° F.

4.4.4.9 High pressure, external. The battery assembly shall be mounted in a test fixture as shown in Figure 4 with the arming wire installed in the piston assembly. A hydrostatic pressure of 1250 ±50 psig shall be applied to the top of the battery case for 5 minutes minimum. Battery electrolyte Type A shall be used as the pressurizing medium. The pressure shall be removed and the battery assembly dried. A low pressure air supply may be used for drying. A voltmeter shall then be connected across the pins of the battery connector. The battery shall be inverted and the arming wire removed. Output potential of the battery shall be 0 volts.

4.4.4.10 Internal leakage. Purge the battery by backfilling from a vacuum of 25 inches of Hg minimum with a volumetric mixture containing 10% minimum helium with the balance being dry nitrogen. Close the battery and insert the arming wire or equivalent. The battery shall be placed in a vacuum chamber and vacuum helium leak tested. It shall exhibit a leak rate less than 1×10^{-5} atm cc/sec. Upon passing this test, the battery shall not be opened until operated.

4.4.4.11 Arming wire pull test. The battery assembly shall be subjected to test (a) and thereafter be subjected to test (b). The battery assembly shall be readied for each test by cutting through the 22-1/4 inch leg of the arming wire in the red cutoff area near the 11/32" x 45° wire bend.

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(a) The battery assembly shall be mounted in accordance with Figure 2. There shall be no separation of the arming wire from the battery assembly when a 20 ±1 pound tension load is suddenly applied at an angle of zero degrees with the top face of the battery assembly.

(b) The battery assembly previously subjected to 4.4.4.11(a) shall be mounted in accordance with Figure 3. The arming wire shall separate from the battery assembly when an 80 ±2 pound tension load is applied suddenly at an angle 60 degrees from the top face of the battery assembly and the cap shall open.

4.4.4.12 Accelerated storage life, packaged and packed condition. The battery assembly, in the packaged and packed condition, shall be subjected to an environmental temperature of +140° F +5° -0° F, at ambient atmospheric conditions, for a period of 30 days. Thereafter, the battery assembly shall be operated in accordance with and meet the requirements of 4.4.4.14.

4.4.4.13 Accelerated storage life, ready condition. The battery assembly, in the ready condition, shall be subjected to an environmental temperature of +140° F +5° -0° F, at ambient atmospheric conditions, for a period of 14 days. Thereafter, the battery assembly shall be operated in accordance with and meet the requirements of 4.4.4.14.

4.4.4.14 Low temperature, low salinity operation. The battery assembly, after being conditioned for 1 hour minimum at +2° F ±2° F, shall be mounted in a test fixture similar to that shown in Figure 5. The battery arming wire shall be removed. The electrical test circuit shall be in accordance with Figure 6. The test fixture and electrolyte, Type A, shall have been previously conditioned to +33° F ±1° F. Operation of the control valve shall release the electrolyte, resulting in a head of 12 inches of solution above the battery. Within 1.0 second after exposure to the electrolyte, the battery assembly shall supply 1.7 volts minimum across a 0.40 ±0.05 ohm resistive load for a minimum of 10 seconds. Polarity of the battery assembly shall conform to Figure 1. Electrolyte is defined in 6.3.9.

4.4.4.15 High temperature, high salinity operation. The battery assembly, after being conditioned for 1 hour minimum at +140° F ±5° F, shall be mounted in a test fixture similar to that shown in Figure 5. The battery arming wire shall be removed. The electrical test circuit shall be in accordance with Figure 6. The test fixture and electrolyte, Type B, shall have been previously conditioned to +98° F ±5° F. Operation of the control valve shall release the electrolyte, resulting in a head of 12 inches of solution above the battery. Within 0.5 second after exposure to the electrolyte, the battery assembly shall supply 2 volts minimum across a 0.40 ±0.05 ohm resistive load for a minimum of 10 seconds. Polarity of the battery assembly shall conform to Figure 1. Electrolyte is defined in 6.3.9.

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4.4.5 Test data sheets. All test data shall be recorded on a data sheet by the supplier. Data sheets and oscillographic records shall be maintained as a part of the inspection records. Each trace on the oscillographic record shall be identified as to function and calibration. In addition to the data requirements of 4.5.9, the data sheets shall contain, but are not limited to, the following information:

FIRST ARTICLE INSPECTION, BATTERY TEST DATA

(a) Date_____

(b) Battery Serial No._____

(c) Instrument Calibration Date_____

(d) High Pressure External_____psig

(e) Internal Leakage_____atm cc/sec.

(f) Arming Wire Pull Test_____

(g) Operation Test, Low Temp._____High Temp._____

Battery Assembly_____Degrees F

Electrolyte_____Degrees F

Electrolyte_____Percent Salinity

Output

Time in seconds:	0.1	0.2	0.3	0.4	0.5	1.0	2.0	3.0	5.0	10.0
Voltage:	___	___	___	___	___	___	___	___	___	____

(h) Polarity: Pin A_____, Pin B_____

4.4.6 Determination of defect or failure. Failure of a first article battery assembly to successfully pass any of the tests in 4.4 through 4.4.4.15 shall be cause for rejection of the first article sample.

4.5 Quality conformance inspection. Battery assemblies and cells produced under the contract or purchase order shall be subjected to the following tests and examinations:

4.5.1 Tests and examinations

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4.5.1.1 Battery assembly

(a) High pressure, internal

(b) Insulation resistance, pin-to-pin

(c) Insulation resistance, pin-to-case

(d) Circuit resistance, pin-to-pin

(e) Internal leakage

(f) Configuration

(g) Arming wire pull test

(h) Low temperature, low salinity operation

4.5.1.2 Battery cell

(a) Insulation resistance

(b) Low temperature, low salinity operation

4.5.2 Each battery assembly in the lot shall be subjected to the tests detailed in 4.5.7.1 through 4.5.7.5.

4.5.3 A sample quantity of battery assemblies shall be selected using the plans and procedures set forth in MIL-STD-105, Inspection Level II, AQL value of 1.0, and shall be subjected to the tests defined in 4.5.7.6 through 4.5.7.8.

4.5.4 A sample quantity of battery cells shall be selected using the plans and procedures set forth in MIL-STD-105, Inspection Level II, AQL value of 0.15, and shall be subjected to the tests defined in 4.5.8.1 and 4.5.8.2.

4.5.5 Test conditions. Unless otherwise specified, the battery assemblies and cells shall be subjected to the quality conformance tests under the following conditions:

(a) Temperature: Room ambient +32° F to +98° F

(b) Altitude: Normal ground atmospheric pressure

(c) Vibration: None

(d) Humidity: Room ambient

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4.5.6 Test equipment and setup. The test setup shall be in accordance with Figure 6.

(a) Oscillographic recorder. An oscillographic recorder shall be used to record the battery data indicated in Figure 6. The recorder shall have the following characteristics:

(1) 2 channels minimum

(2) Permanent recording

(3) Frequency response: DC to 40 cycles per second (cps) minimum

(4) Sensitivity: 0.10 volt per inch minimum

(5) Time base: 100 ms markers

(6) Chart speed: Variable in steps, 1 to 10 inches per second

(7) Recording paper or film: 2-inch width minimum

(8) Input impedance: 1000 ohms minimum

(b) Leak detector. Sensitivity of 1×10^{-4} atm cc/sec helium into air.

4.5.7 Test procedures, battery assembly

4.5.7.1 High pressure, internal. Prior to final assembly, each battery case and each connector shall be temporarily assembled and subjected to a hydrostatic pressure of 1250 ±50 psig for 5 minutes minimum. There shall be no visible evidence of leakage or permanent deformation.

4.5.7.2 Insulation resistance, pin-to-pin. With the arming wire installed, the resistance between pins A and B of the battery connector shall exceed 20 megohms when measured at a potential of 50 ±10 volts direct current.

4.5.7.3 Insulation resistance, pin-to-case. With the arming wire removed, the resistance between both pins A and B of the battery connector and the battery case shall exceed 20 megohms when measured at a potential of 50 ±10 volts direct current.

4.5.7.4 Circuit resistance, pin-to-pin. With the arming wire removed, the resistance between pins A and B of the battery connector shall be 470 K ± 94 K ohms.

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4.5.7.5 Internal leakage. Purge the battery by backfilling from a vacuum of 25 inches of Hg minimum with a volumetric mixture containing 10% minimum helium with the balance being dry nitrogen. Close the battery and insert the arming wire or equivalent. The battery shall be placed in a vacuum chamber and vacuum helium leak tested. It shall exhibit a leak rate less than 1 x 10^{-5} atm cc/sec. Upon passing this test, the battery shall not be opened until operated.

4.5.7.6 Configuration (Refer to NAVORD Dwg 2526883)

(a) Dimensions

(1) 2.121 ±0.002 dia

(2) 1.946 ±0.003 dia

(3) The 2.121 dia shall be concentric to (-A-) dia within 0.005 TIR

(4) The 1.946 dia shall be concentric to (-B-) dia within 0.002 TIR

(5) The surface roughness on the 1.946 dia shall not exceed 32 microinches

(6) 2.395 ±0.010 length

(b) Marking. The marking shall be in accordance with the drawing.

4.5.7.7 Arming wire pull test. The battery assembly shall be subjected to test (a) and thereafter be subjected to test (b). The battery assembly shall be readied for the test by cutting through the 22-1/4 inch leg of the arming wire in the red cutoff area near the 11/32" x 45° wire bend.

(a) The battery assembly shall be mounted in accordance with Figure 2. There shall be no separation of the arming wire from the battery assembly when a 20 ±1 pound tension load is suddenly applied at an angle of zero degrees with the top face of the battery assembly.

(b) The battery assembly previously subjected to 4.5.7.7(a) shall be mounted in accordance with Figure 3. The arming wire shall separate from the battery assembly when an 80 ±2 pound tension load is applied suddenly at an angle 60 degrees from the top face of the battery assembly and the cap shall open.

4.5.7.8 Low temperature, low salinity operation. The battery assembly, after being conditioned for 1 hour minimum at +2° F ±2° F, shall be mounted in a test fixture similar to that shown in Figure 5. The battery

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arming wire shall be removed. The electrical test circuit shall be in accordance with Figure 6. The test fixture and electrolyte, Type A, shall have been previously conditioned to +33° F ±1° F. Operation of the control valve shall release the electrolyte, resulting in a head of 12 inches of solution above the battery. Within 1.0 second after exposure to the electrolyte, the battery assembly shall supply 1.7 volts minimum across a 0.40 ±0.05 ohm resistive load for a minimum of 10 seconds. Polarity of the battery assembly shall conform to Figure 1. Electrolyte is defined in 6.3.9.

4.5.8 Test procedures, battery cell

4.5.8.1 Insulation resistance. The insulation resistance of the battery cell between the two leads, shall be greater than 20 megohms when measured at a potential of 50 ±10 volts direct current.

4.5.8.2 Low temperature, low salinity operation. The battery cell, after being conditioned for 1 hour minimum at +2° F ±2° F, shall be mounted in a test fixture similar to that shown in Figure 7. The electrical test circuit shall be in accordance with Figure 6. The test fixture and electrolyte, Type A, shall have been previously conditioned to +33° F ±1° F. Operation of the control valve shall release the electrolyte, resulting in a head of 12 inches of solution above the battery. Within 1.0 second after exposure to the electrolyte, the cell shall supply 2.5 volts minimum across a 0.40 ±0.05 ohm resistive load for a minimum of 10 seconds. Polarity of the cell shall conform to Figure 7. Electrolyte is defined in 6.3.9.

4.5.9 Test data sheets. All test data shall be recorded on a data sheet by the supplier. Data sheets and oscillographic records shall be maintained as a part of the inspection records. Each trace on the oscillographic record shall be identified as to function and calibration. Data sheets shall contain, but are not limited to, the following information:

QUALITY CONFORMANCE INSPECTION, BATTERY TEST DATA

Battery Assembly

(a) Date_____

(b) Serial No._____

(c) Instrument Calibration Data_____

(d) Internal pressure_____psig_____minutes

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(e) Resistance

Pin-to-pin (Arming wire installed)_____

Pin-to-case (Arming wire removed)_____

Pin-to-pin (Arming wire removed)_____

(f) Configuration

(g) Leak test_____atm cc/sec

(h) Arming wire pull test_____

(i) Operational test

Battery Temp._____Degrees F

Electrolyte Temp._____Degrees F

Electrolyte_____Percent Salinity

Time in seconds: 0.1 0.2 0.3 0.4 0.5 1.0 2.0 3.0 5.0 10.0

Output voltage: ___ ___ ___ ___ ___ ___ ___ ___ ___ ___

(j) Polarity: Pin A_____, Pin B_____.

Battery Cell

(a) Date_____

(b) Instrument calibration date_____

(c) Operational Test

Cell Temp._____Degrees F

Electrolyte Temp._____Degrees F

Electrolyte_____Percent Salinity

Time in seconds: 0.1 0.2 0.3 0.4 0.5 1.0 2.0 3.0 5.0 10.0

Output voltage: ___ ___ ___ ___ ___ ___ ___ ___ ___ ___

(d) Polarity_____

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4.5.10 Determination of defect or failure

4.5.10.1 Battery assembly. Failure of a battery assembly to successfully pass any of the tests in 4.5.7.1 through 4.5.7.5 shall be cause for rejection of the battery. Failure of a battery assembly to successfully pass the tests in 4.5.7.6 through 4.5.7.8 shall constitute a defect. Acceptance of the lot shall be in accordance with MIL-STD-105 Level II, AQL 1.0.

4.5.10.2 Battery cell. Failure of a battery cell to successfully pass the tests in 4.5.8.1 or 4.5.8.2 shall constitute a defect. Acceptance of all battery cells in that lot shall be in accordance with MIL-STD-105 Level II, AQL 0.15.

4.6 Periodic production inspection. Periodic production samples of the battery assembly shall be selected in accordance with 4.3.3. The samples shall be tested in accordance with the following plan:

(a) The sample of 3 units from the first accepted battery lot will be called the first sample and will undergo the tests of Table IX for sample 1. One unit will be tested to the first sequence and 2 units will be tested to the second sequence.

(b) The sample of 3 units from the second and third accepted battery lots will be called the second and third samples and will undergo the tests of Table IX for samples 2 and 3 respectively. The sequence of tests and number of units are shown in Table IX.

(c) The next sample of 3 units from the fourth accepted battery lot will be called the fourth sample and will undergo the tests of Table IX for sample 4, etc.

TABLE IX

PERIODIC TEST SEQUENCE

Tests	Sample and Test Sequence						
	Sample Lot No.	1, 4, Etc.		2, 5, Etc.		3, 6, Etc.	
	No. of Units	1	2	1	2	1	2
Inspection		1	1	1	1	1	1
Vibration, Ready Condition					2*		2*
Acceleration, Ready Condition							3*
Shock, Ready Condition			2				4*
Temperature and Humidity			3		3		5
Marine Atmosphere			4				

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TABLE IX (Cont'd)

PERIODIC TEST SEQUENCE

Tests	Sample and Test Sequence					
	Sample Lot No.	1, 4, Etc.		2, 5, Etc.		3, 6 Etc.	
	No. of Units	1	2	1	2	1	2
Accelerated Storage Life Test, Ready Condition		2		2		2	
Low Temperature, Low Salinity Operation		3	5	3		3	6
High Temperature, High Salinity Operation					4		

*Visual check after testing

4.6.1 Test conditions. Unless otherwise specified, the battery assembly shall be subjected to the periodic tests under the following conditions:

(a) Temperature: Room ambient +32° F to +98° F

(b) Altitude: Normal ground atmospheric pressure

(c) Vibration: None

(d) Humidity: Room ambient

4.6.2 Test equipment and setup. The test equipment and setup for the periodic tests shall be in accordance with 4.4.3.

4.6.3 Test procedures

4.6.3.1 Inspection. The battery assemblies shall be subjected to the inspections detailed in 4.5.7.1 through 4.5.7.6.

4.6.3.2 Vibration, ready condition. Vibration testing of the battery assembly, in the ready condition, shall be performed in accordance with 4.4.4.2.

4.6.3.3 Acceleration, ready condition. Acceleration testing of the battery assembly, in the ready condition, shall be performed in accordance with 4.4.4.3.

4.6.3.4 Shock, ready condition. Shock testing of the battery assembly, in the ready condition, shall be performed in accordance with 4.4.4.4.

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4.6.3.5 Temperature and humidity, nonoperating condition. Temperature and humidity testing of the battery assembly, in the nonoperating condition, shall be performed in accordance with 4.4.4.5.

4.6.3.6 Marine atmosphere. The battery assembly shall be subjected to the marine atmosphere test in accordance with 4.4.4.6.

4.6.3.7 Accelerated storage life, ready condition. Accelerated storage life testing of the battery assembly, in the ready condition, shall be performed in accordance with 4.4.4.13.

4.6.3.8 Low temperature, low salinity operation. Low temperature, low salinity operation of the battery assembly shall be performed in accordance with 4.4.4.14.

4.6.3.9 High temperature, high salinity operation. High temperature, high salinity operation of the battery assembly shall be performed in accordance with 4.4.4.15.

4.6.4 Test data. Test data shall be recorded in accordance with 4.4.5, as appropriate, and maintained as a part of the inspection record.

4.6.5 Determination of defect or failure. Failure of the battery assembly to successfully pass the tests in 4.6.3.1 through 4.6.3.9 shall be cause for withholding acceptance of all units on hand.

5. PREPARATION FOR DELIVERY

5.1 The battery assembly shall be packed and packaged in accordance with instructions listed on NAVORD DL 1330510.

5.2 Marking. In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The battery assembly is intended for use in a torpedo to provide electrical energy to initiate the starting sequence of the torpedo propulsion system.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification

(b) Assigned activity for first article tests

(c) Any exceptions to this specification

(d) Inspection data required

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6.3 Definitions

6.3.1 Ready. The condition of the battery assembly when fully assembled in its operational chamber or equivalent, with the arming wire in place.

6.3.2 Operating. The condition of the battery assembly when fully assembled in its operational chamber or equivalent with the arming wire removed and the electrolyte added.

6.3.3 Nonoperating. The condition of the battery assembly when stored, transported, or handled. Tests on the battery assembly, in the nonoperating condition, shall be performed without packaging or packing.

6.3.4 Vibration. Vibration is a sinusoidal motion defined by amplitude and frequency. The amplitude may refer to displacement or to acceleration. The displacement is given in inches peak-to-peak or the corresponding maximum acceleration in gravitational units. The frequency is given in cycles per second.

6.3.5 Acceleration (steady state). Acceleration (steady state) is the time rate of change of velocity of a body resulting from the application of a sustained external force with a uniform buildup to peak in more than 0.010 second. Acceleration is given in gravitational units.

6.3.6 Shock. Shock is an abrupt change in the motion of a body resulting from the sudden application of a large external force of short duration. Shock, in general, can be described as a buildup to 90 percent of peak force in less than 0.010 second. Shock is described as a pulse in units of gravitational acceleration, duration in milliseconds and shock wave form equivalent to one half sine wave. Duration is defined as the interval between the point of zero amplitude, from which the shock wave form builds up, to its decay to zero amplitude.

6.3.7 Temperature. The temperature of the environment to which the battery assembly is exposed, is expressed in degrees Fahrenheit.

6.3.8 Humidity. Humidity is the relative humidity of the environment to which the battery assembly is exposed, expressed in percent.

6.3.9 Electrolyte. The electrolyte shall consist of tap water and sea salts. Type A solution is defined as a 1.05% ±0.05% solution by weight. Type B solution is defined as a 3.95% ±0.05% solution by weight. Typical sea salts consist of the following chemicals in the proportions listed:

Chemical	Percent by Weight
Na Cl	58.490
$MgCl_2-6H_2O$	26.460
$NaSO_4$	9.750

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Chemical	Percent by Weight
$Ca\ Cl_2\text{-}2H_2O$	2.765
K Cl	1.645
$Na\ HCO_3$	0.477
K Br	0.238
Other	0.175

6.4 Weight. The weight of the battery assembly shall not exceed 1 pound.

Preparing Activity
Navy - OS
Project No. 1355-N044

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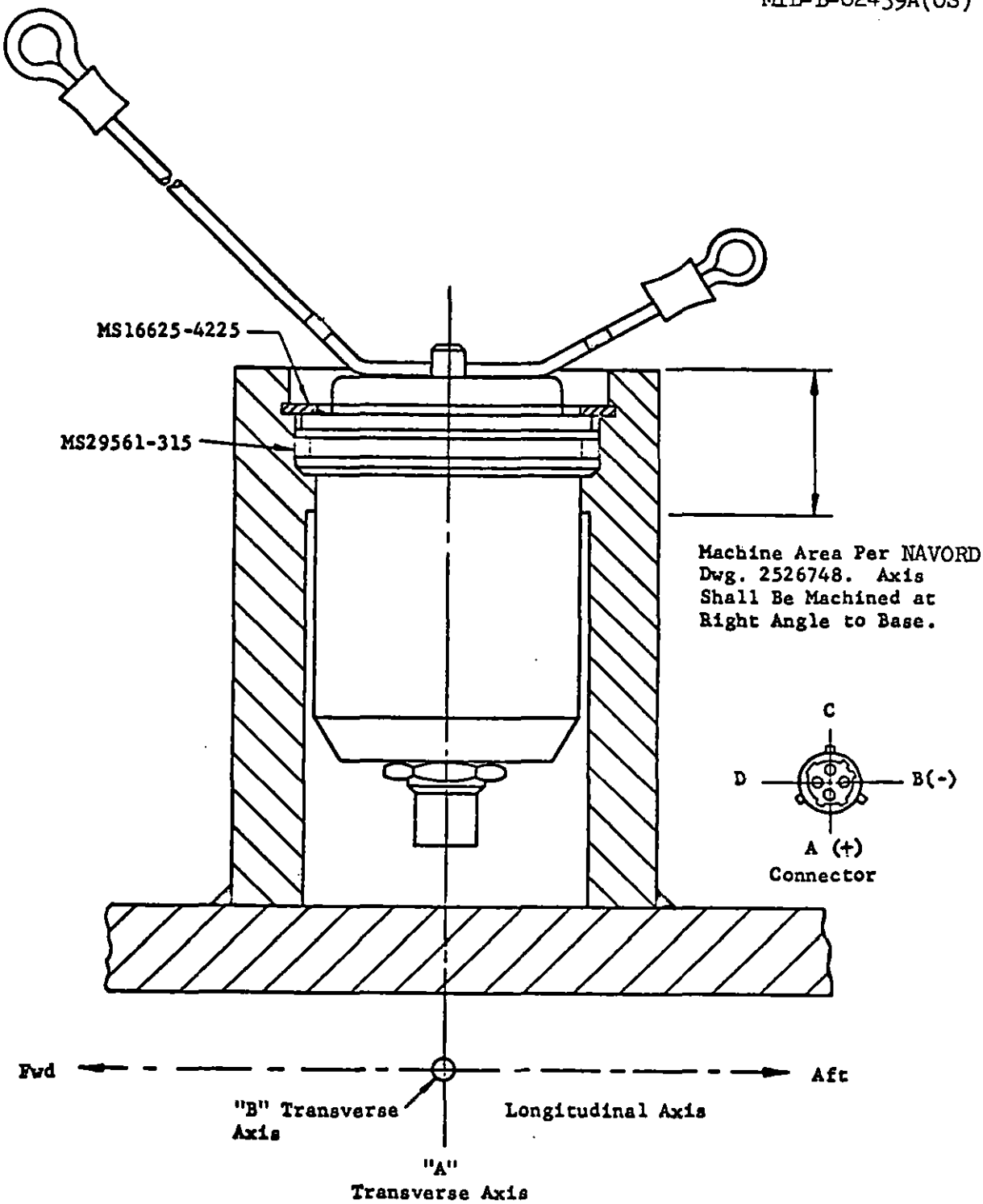
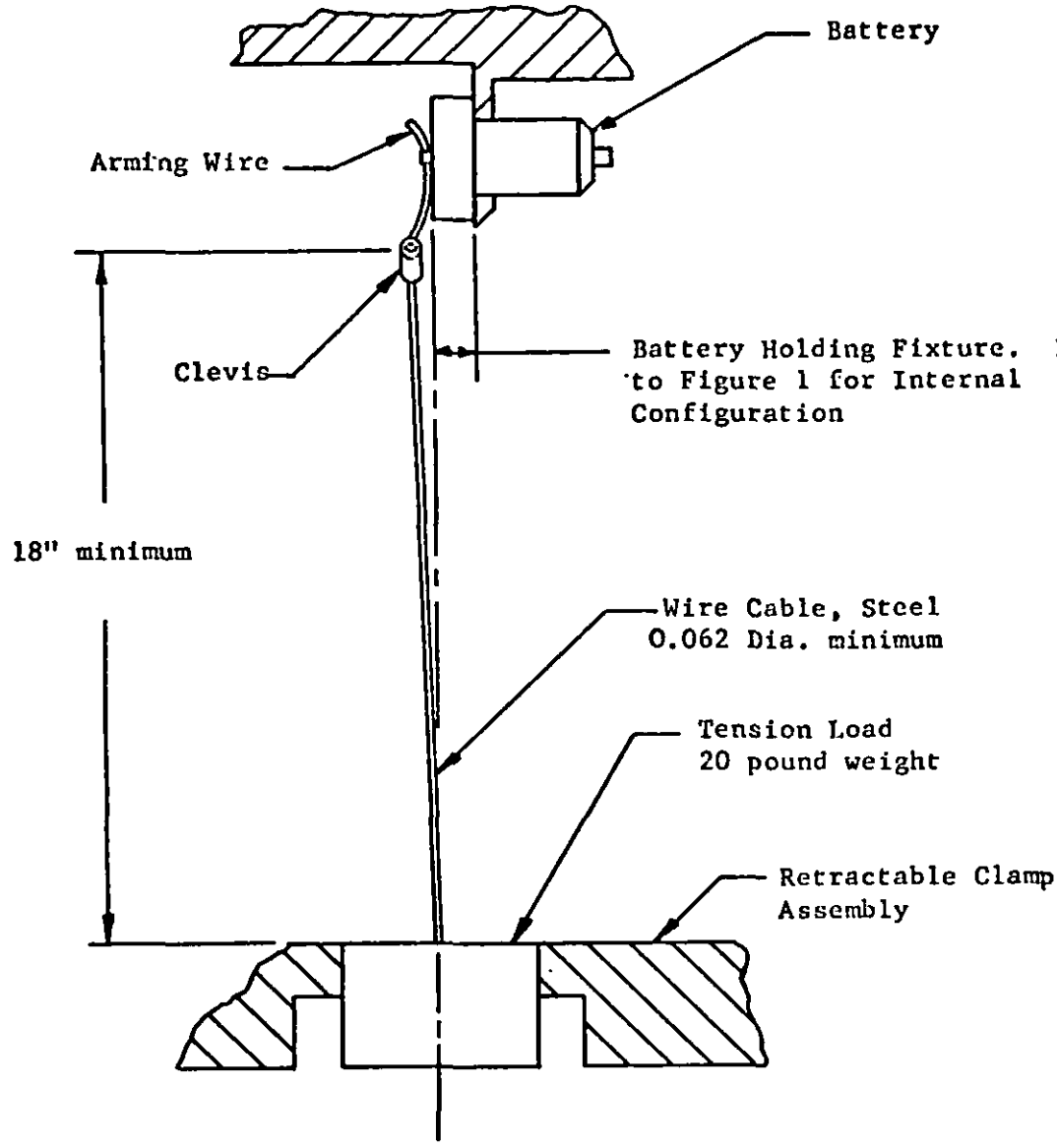


Figure 1. Battery Assembly Support Frame

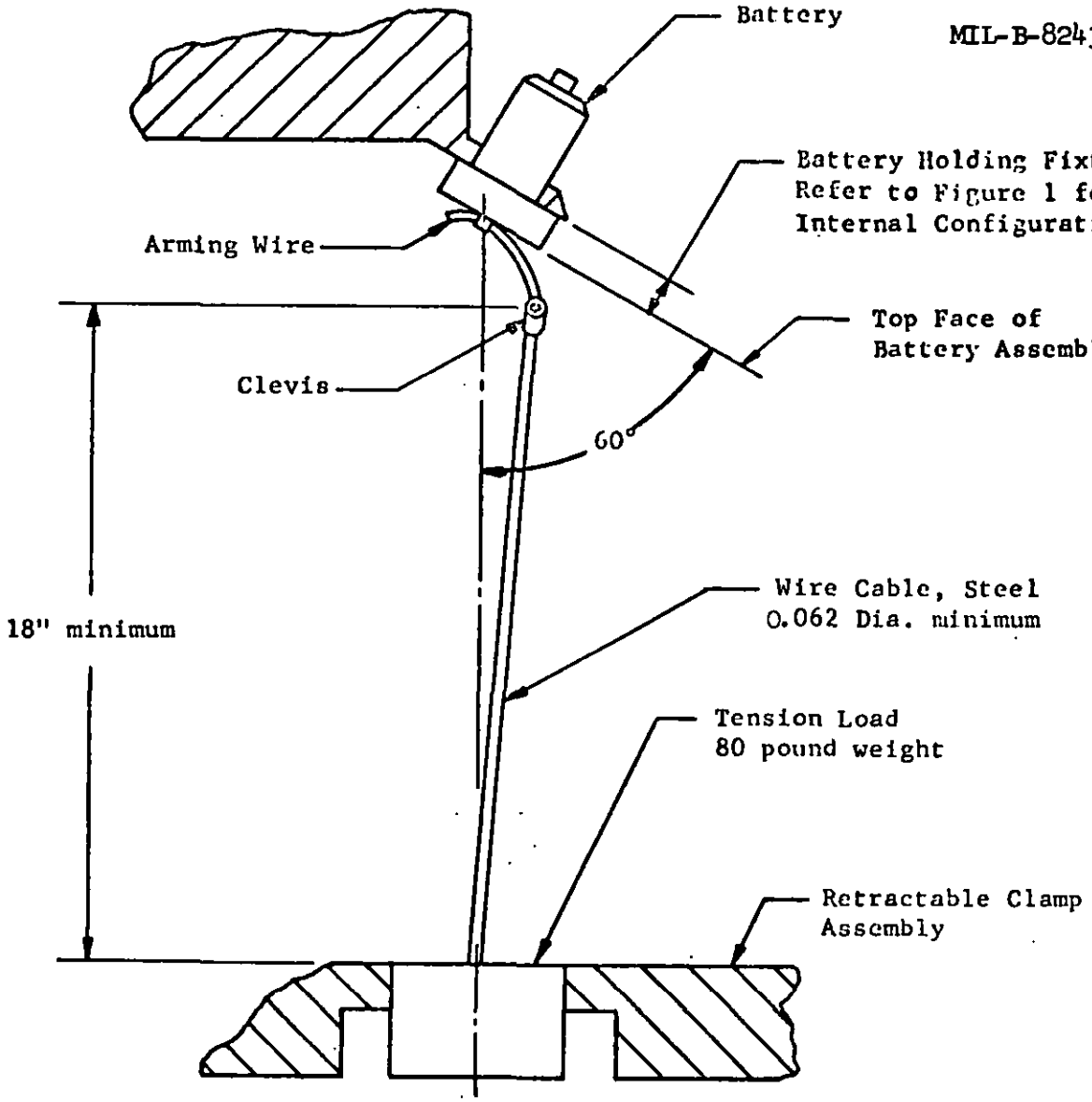
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NOTE: Before the load is released, the arming wire shall be pulled through the hole in the battery piston until the 45 degree bend in the arming wire is snug against the piston. Any slackness in the cable between the arming wire and the tension load shall be removed before the load is released. The load clamping mechanism shall be such that there is an instantaneous retraction of the clamp, giving complete freedom for the load to drop.

Figure 2. Arming Wire Zero Degree Pull Test Diagram

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NOTE: Before the load is released, the arming wire shall be pulled through the hole in the battery piston until the 45 degree bend in the arming wire is snug against the piston. Any slackness in the cable between the arming wire and the tension load shall be removed before the load is released. The load clamping mechanism shall be such that there is an instantaneous retraction of the clamp, giving complete freedom for the load to drop.

Figure 3. Arming Wire 60 Degree Pull Test Diagram

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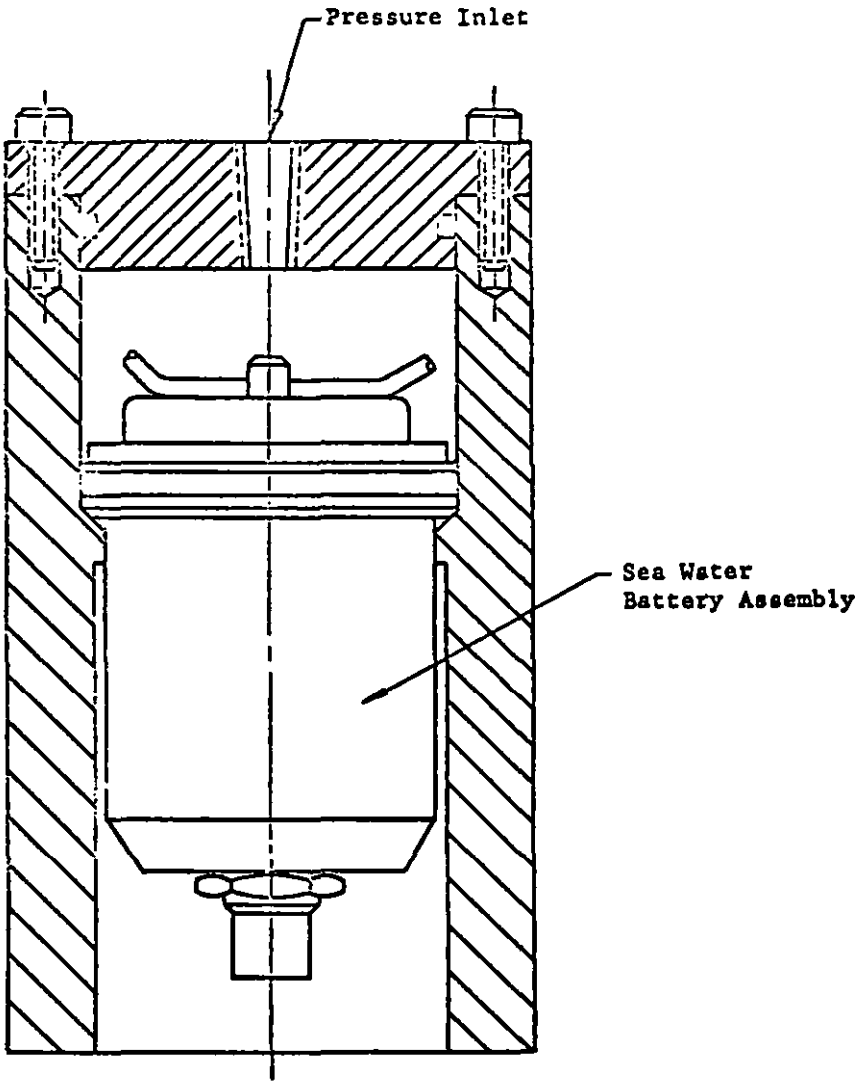


Figure 4. Pressure Test Fixture

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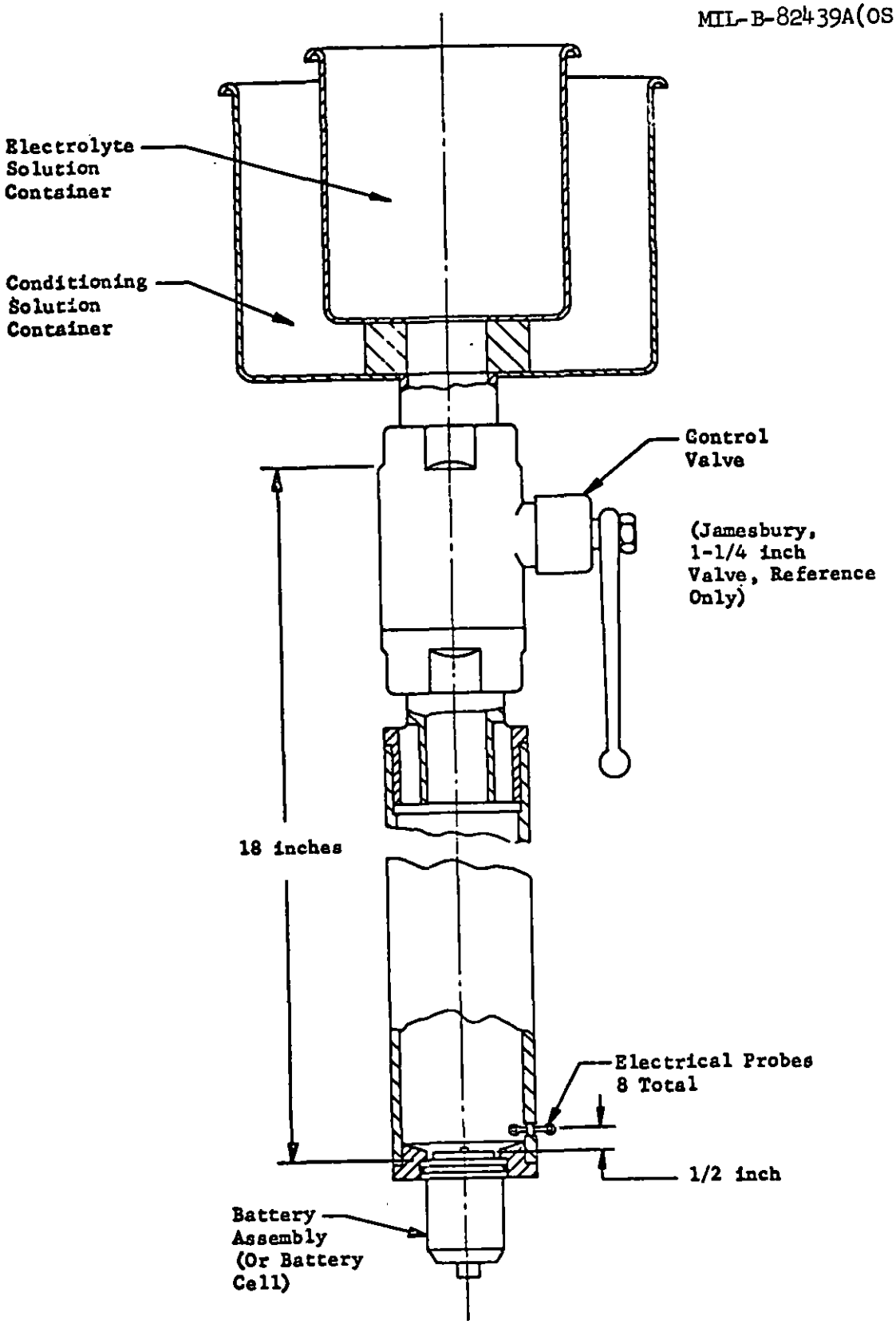


Figure 5. Operational Test Fixture

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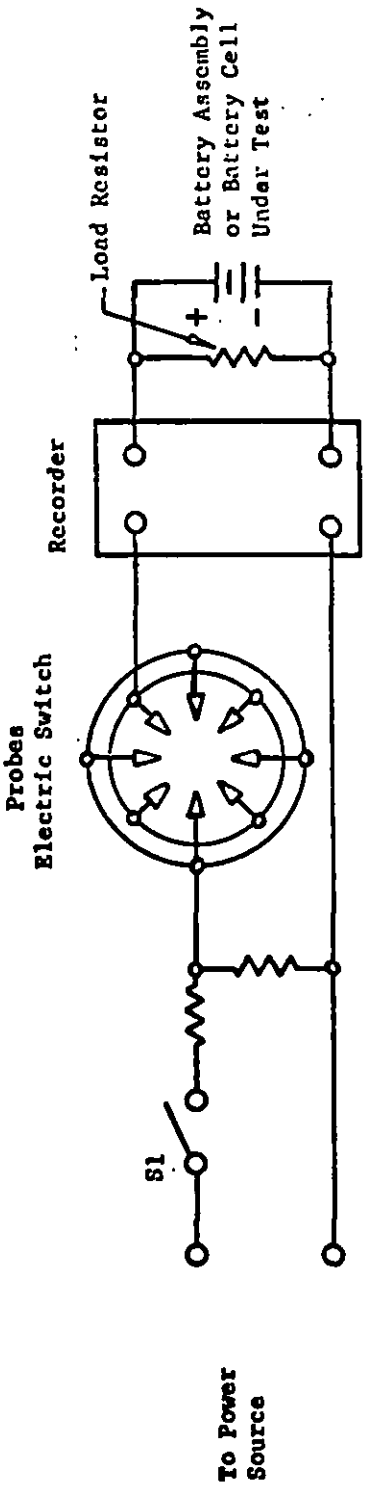
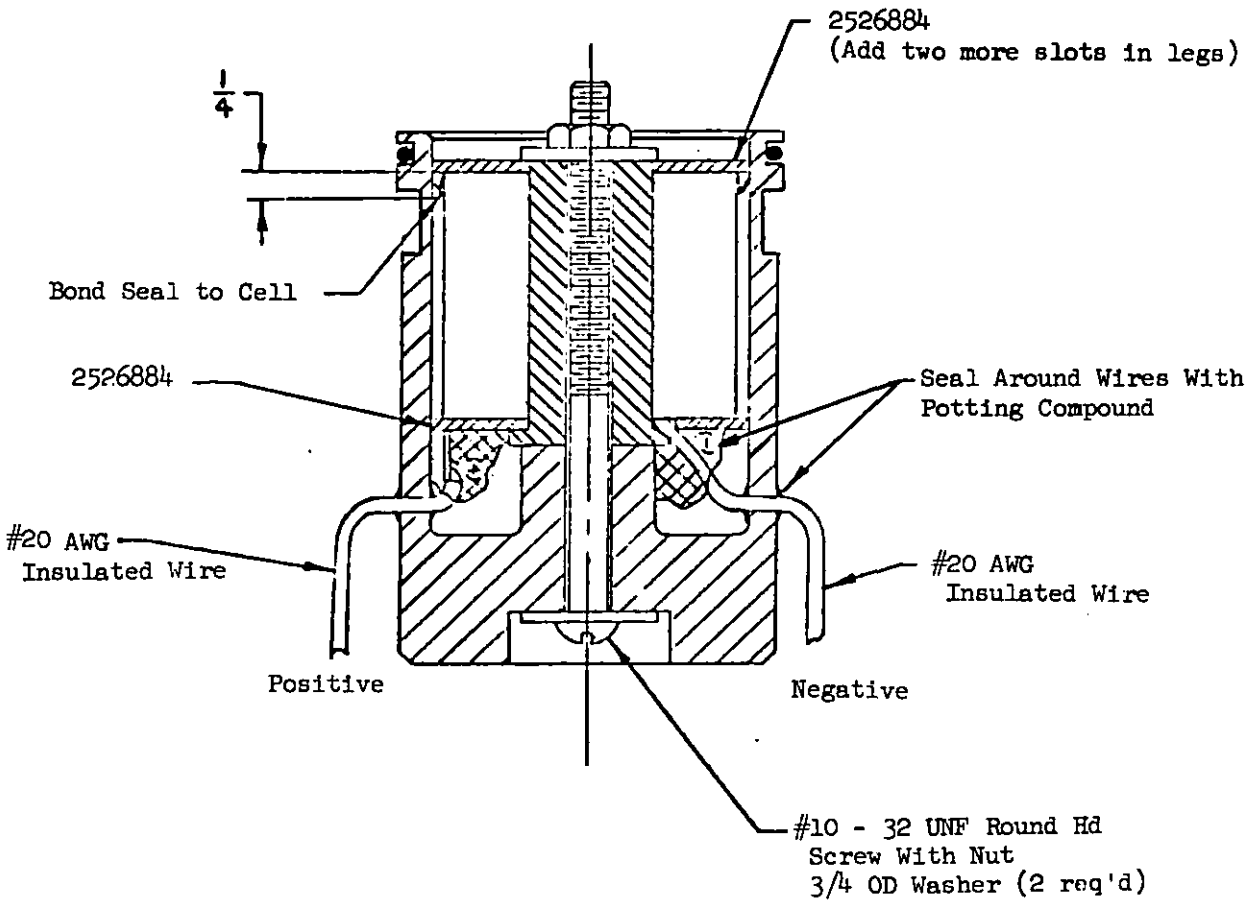


Figure 6. Electrical Test Circuit

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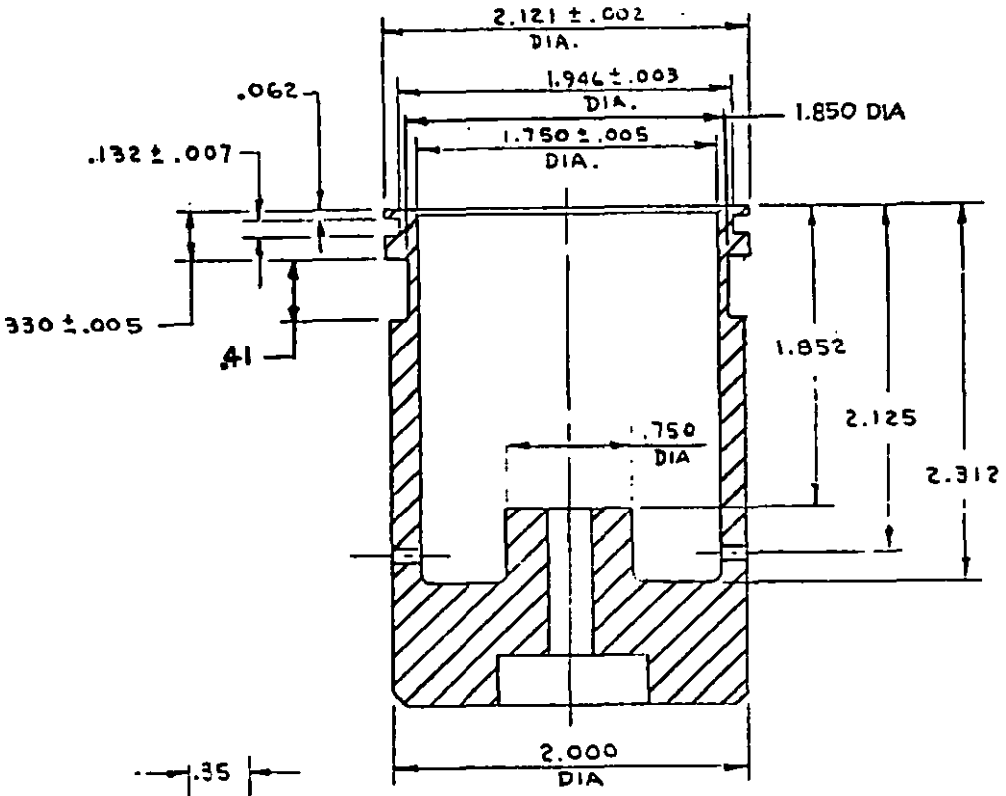
TEST CELL ASSEMBLY



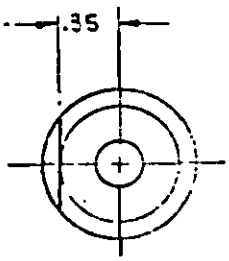
NOTE: See Figure 8 For Details

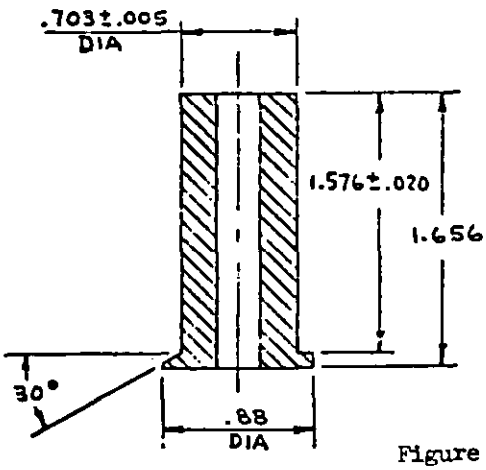
Figure 7. Operational Cell Test Fixture

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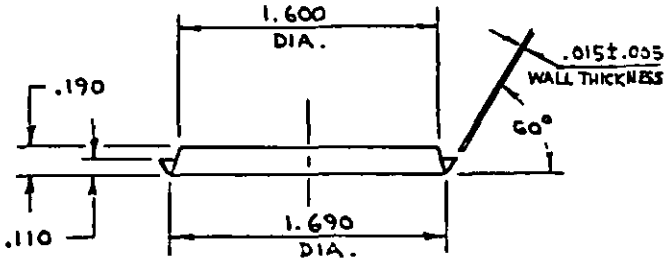


HOUSING





ROD
Mat'l: Non-metallic



SEAL
Mat'l: Neoprene 50±10 Durometer

Figure 8. Cell Fixture Details

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393-088 (S-178)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-B-82439A(OS) BATTERY, SEA WATER

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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